Magnetic Moment Method with the Idea of Magnetic Surface Charge Method

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Magnetic Moment Method (MMM) is well known for its "lightweight" and simplicity of implementation; however, meshing must be carefully treated to obtain accurate results. In this paper, the MMM with the idea of the Magnetic Surface Charge method is proposed that is free from the meshing problem. Accelerator magnets; a C-shaped dipole and a quadrupole magnet, are analyzed, and it is shown that the proposed method outperforms currently available finite-element packages with respect to the CPU time and accuracy of the magnetic field.

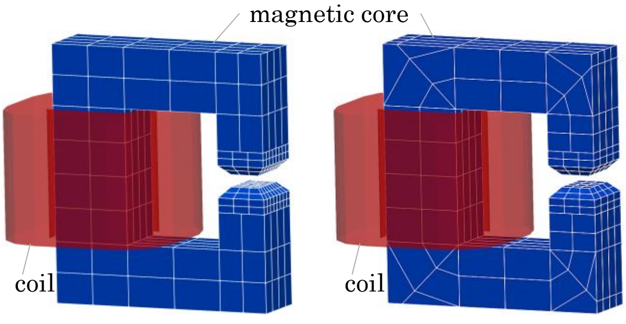
Keywords: electromagnetic field analysis, accelerator magnets, magnetic moment method, magnetic surface charge method, meshing

1. Introduction

Most of the accelerator magnets are designed by means of electromagnetic simulations employing three-dimensional finite element method. OPERA-3D (TOSCA) is one of the major software for modeling accelerator magnets [1), 2)]. The name TOSCA comes from two scalar potentials, namely, reduced potentials and total potentials. The method has the following advantages. (i) Because it is based on the two-scalar-potential formulation, the number of degrees of freedom is small, and the magnetostatic analysis is fast. (ii) Coils are modeled as Biot-Savart sources and do not have to be meshed; therefore, the mesh quality of the magnetic core can easily be controlled.

Another famous software to model accelerator magnets is RADIA [3)–5)], which is freely distributed as Mathematica's add-on by ESRF: the European Synchrotron. The method has the following advantages. (i) Because it is based on the Magnetic Moment Method (hereafter MMM [6),7)]), no air-mesh is required, and no special technique is required to emulate open boundaries. (ii) Because only the magnetic core is meshed, it is fast and "lightweight," namely the size of the data is small. One problem of the MMM, which is also stated on Radia's website [3)], is the meshing problem. Figures 1 show a C-shaped electromagnet subdivided into (a) parallelepiped mesh and (b) mesh parallel to flux lines. If the iron core is subdivided into parallelepiped mesh, as shown in Fig. 1 (a), it is known that an inaccurate magnetic field is obtained. To avoid it, the iron core must be meshed parallel to flux lines, as shown in Fig. 1 (b), however, flux-line distributions must be known before solving the problem.

In this paper, the MMM with the idea of the Magnetic Surface Charge method (hereafter MMM with MSC) is proposed that is free from the above-mentioned meshing problem. Unknown variables in the MMM with MSC are surface magnetic charges on each face of the elements, whereas unknown variables in the MMM are magnetic moment vectors in each element. The negative magnetic point charges are placed at the center of the gravity of the element to guarantee Gauss's law in each element whose magnitudes are equal to the surface integral of each surface charge. By doing so, the bent of the magnetic flux in each element can be reproduced more naturally than the conventional MMM, even with coarse mesh. Accelerator magnets, such as a C-shaped dipole and a quadrupole magnet, are analyzed, and it is shown that the proposed method outperforms currently available finite-element packages with respect to the CPU time and accuracy of the magnetic field.



(a) parallelepiped mesh (b) parallel to flux lines

Fig. 1 Two different types of mesh for MMM.

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2. Formulation

2-1. Conventional Magnetic Moment Method

A magnetic moment $\boldsymbol{M}$ in the element produces magnetic field $\boldsymbol{H}$ at observation point $\boldsymbol{r}$ by taking integrals as (1) where $\boldsymbol{r}'$, is the displacement $\boldsymbol{s}'$ is the surface, $\boldsymbol{V}'$, is the volume of the element, and $\boldsymbol{n}_{s'}$ is the normal vector of each surface[5)].

$$\boldsymbol{H}(\underline{\mathrm{r}}) = \frac{1}{4\pi}\int \frac{(\boldsymbol{r}'-\boldsymbol{r})}{|\boldsymbol{r}'-\boldsymbol{r}|^3}\nabla\cdot\boldsymbol{M}dV' - \frac{1}{4\pi}\oint \frac{(\boldsymbol{r}'-\boldsymbol{r})}{|\boldsymbol{r}'-\boldsymbol{r}|^3}\boldsymbol{M}\cdot\boldsymbol{n}_{s'}dS' \tag{1}$$

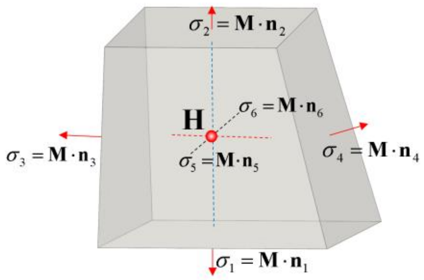


Fig. 2 Schematic of MMM's concept.

Figure 2 shows a schematic the concept of the MMM. Unknown variables in the MMM are magnetic moments $\boldsymbol{M}$ and thus the number of degrees of freedom is three per element. If $\boldsymbol{M}$ is assumed to be uniform inside the element, the first integral term in (1) vanishes; therefore, the matrix representation of the conventional MMM is derived from (2). The magnetic field produced by $\boldsymbol{M}$ is computed with the equivalent surface magnetic charges $\boldsymbol{M}\cdot\boldsymbol{n}_{s'}$. When the point matching method is employed, (2) is evaluated at the center of gravity of the element to construct a matrix equation.

$$\boldsymbol{H}(\boldsymbol{r}) = -\frac{1}{4\pi}\oint\frac{(\boldsymbol{r}'-\boldsymbol{r})}{|\boldsymbol{r}'-\boldsymbol{r}|^3}\boldsymbol{M}\cdot\boldsymbol{n}_{s'}dS' \qquad (2)$$

2-2. Magnetic Moment Method with the Idea of Surface Magnetic Charge Method

Figures 3 show schematic of the concept of the MMM with MSC. Hexahedral elements are subdivided into six pyramids and unknown variables are surface magnetic charge densities σ_n on each pyramid's bottom surface. Negative magnetic point charges m_n whose magnitudes are equal to the surface integral of the σ_n are placed at the vertex of each pyramid so that the total charges in each element are zeros in total. The number of degrees of freedom is six per element which is double the MMM. The magnetic fields H are evaluated at mid-points between the center of the surface and the center of gravity. The first integral term in (1) corresponds to the negative magnetic point charges and the second term to surface magnetic charges. The permeability is assumed to be constant, whereas the magnetic flux is not uniform and can bend in the element.

Figure 4 shows a simple two-dimensional magnetostatic problem. Two permanent magnets with the remanence of 0.2 T are attached to two adjacent edges of the iron core whose relative permeability is

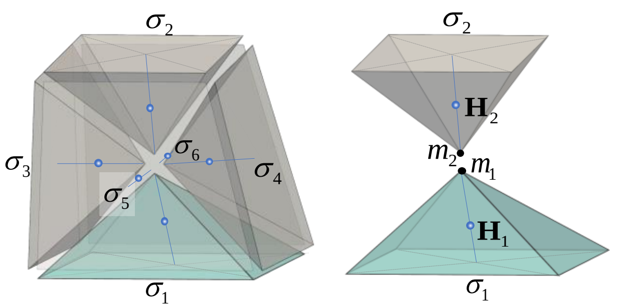
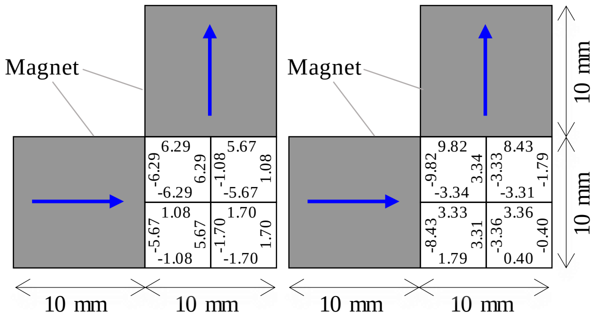


Fig. 3 Schematic of MMM with MSC's concept.



(a) MMM (b) MMM with MSC

Fig. 4 Simple two-dimensional static problem indicating the existence of the theoretical limit that can be achieved by the MMM.

1000. The iron core is subdivided into four squares. The obtained equivalent surface magnetic charge densities σ [Wb/m^2] are shown in Fig. 4. If the permeability is uniform in the iron core, the equivalent magnetic charge densities on the inner surfaces of two adjacent elements must theoretically be zero in total. For example, in the MMM, at the bottom edge of the upper left square, σ = -6.29 Wb/m^2, and at the top edge of the lower left square, σ = 1.08 Wb/m^2, thus, they do not cancel each other. In the MMM with MSC, at the bottom edge of the upper left square σ = -3.34 Wb/m^2, and at the top edge of the lower left square, σ = 3.33 Wb/m^2, they are almost zero in total. This indicates that there is a theoretical limit to the accuracy that can be achieved with the MMM whereas the MMM with MSC is free from it.

3. Numerical results

3-1. T-Shaped magnet

Figures 5 show the T-shaped magnet; a typical example of which the MMM becomes inaccurate. A cuboid magnet is attached to a T-shaped magnetic core. The relative permeability of the magnetic core is 1000. and composed of four cubes with an edge length of 1 mm. The size of the permanent magnet is 1 mm × 1 mm × 2 mm and the and its remanence is 0.2 T The surface magnetic field distributions obtained by the MMM and the MMM with MSC are shown in Fig. 5. Upper figures are when the subdivision number N = 1 and lower figures are with N = 3. The result obtained by FEM with the 5 million mesh is used as the reference solution. Figure 6 shows the convergence of the surface magnetic field at the observation point P in increasing N. Even with N = 3, accurate results within sub-percent error can be obtained by MMM with MSC whereas the convergence of the MMM is slow and not simply monotonic.

3-2. C-Shaped magnet

A C-shaped magnet is a typical accelerator magnet, and an example model of RADIA is provided by ESRF[3]. Figures 7 show the three types of mesh of the C-shaped magnet: (a) parallelepiped coarse mesh, (b) parallelepiped fine mesh, and (c) mesh along flux lines.

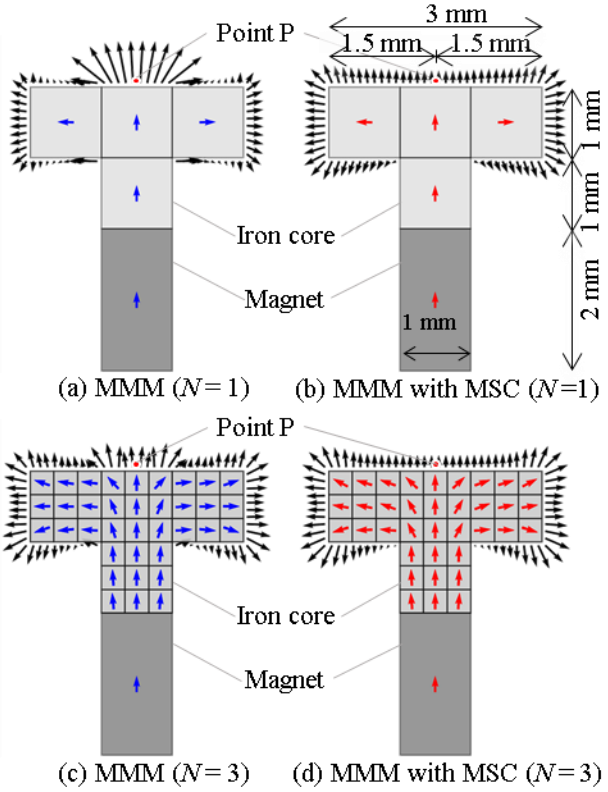


Fig. 5 T-shaped magnet example.

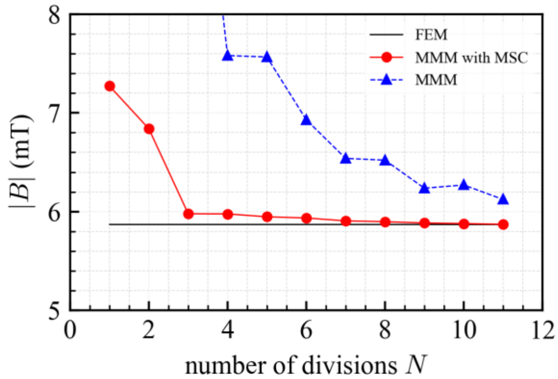


Fig. 6 Surface magnetic field at the point P.

The B-H curve of the magnetic core is shown in Figure 8.

The size of the magnetic core is 182.5 mm × 210 mm. The iron core gap is 10 mm and the magnetic field at the gap center is evaluated with various coil currents from 1000 to 10000 AT. The result obtained by OPERA-3D is used as the reference solution. 8-layer IABC [8), 9)] was required to obtain a self-consistent open boundary solution in OPERA-3D. Figures 9 show the magnetic field at the center of the gap and those errors with three types of mesh. Even with mesh type (a), the error of the magnetic field obtained by the MMM with MSC is about 4% at maximum. Both with mesh type (b) and (c), the error of the magnetic field obtained by the MMM with MSC is within sub percent, which indicates that the MMM with MSC is free from the meshing problem discussed in the introduction. The computing time for analyzing the C-shaped magnet with Intel Core i7-9700K CPU at 3.60 GHz is shown in Table 1. Although the computing time for the MMM is shorter than the MMM with MSC when the number of elements is the same, the results obtained by the MMM with MSC is more accurate especially when the mesh is coarse.

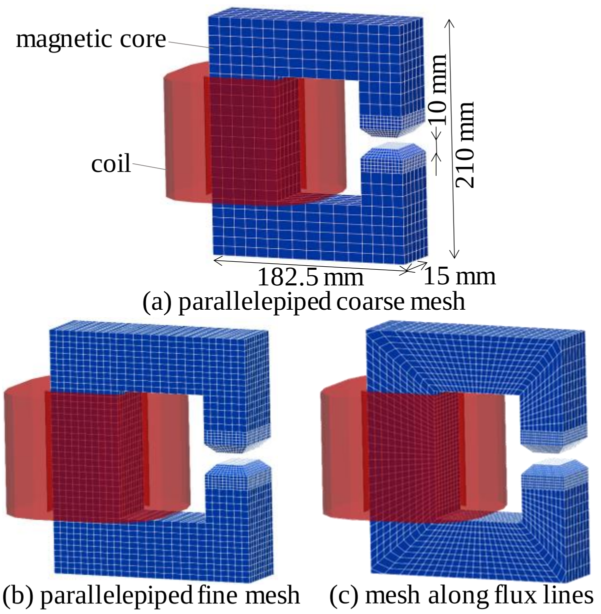


Fig. 7 Various mesh types of the C-shaped magnet.

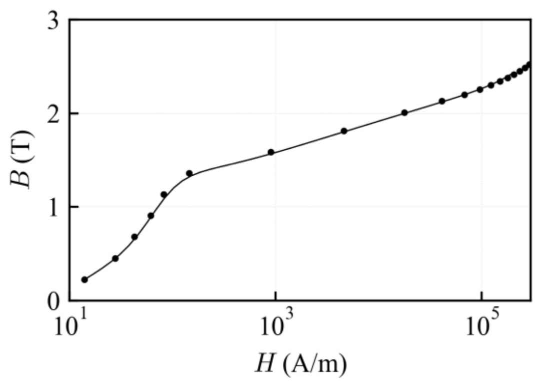


Fig. 8 B-H curve of the iron core.

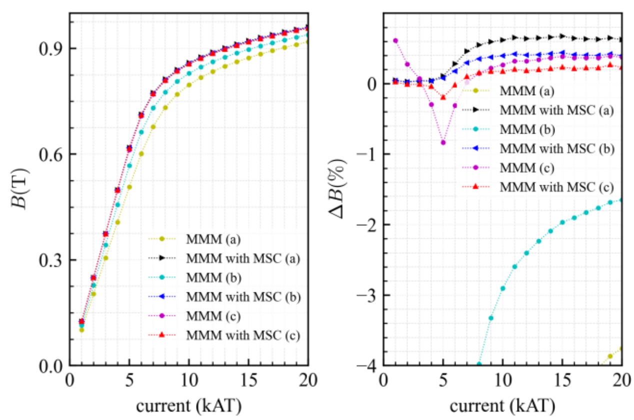


Fig. 9 Magnetic field and its errors at the gap center.

Table 1 CPU time for analysis of C-shaped magnet.

elements		method	current (AT)	CPU time (sec) mesh	CPU time (sec) solver
16,009,450		Opera3D	1000	6506	1885
			20000	6506	12231
(a)	1,200	MMM	1000	< 1	4
			20000	< 1	13
		MMM with MSC	1000	< 1	49
			20000	< 1	64
(b)	3,150	MMM	1000	1.2	29
			20000	1.2	152
		MM with MSC	1000	1.2	919
			20000	1.2	986
(c)	4,008	MMM	1000	1.5	55
			20000	1.5	261
		MMM with MSC	1000	1.5	1955
			20000	1.5	5428

3-3. Quadrupole magnet

A quadrupole magnet is also a typical accelerator magnet, and an example model of RADIA is provided by ESRF[3]. Figures 10 show the mesh of the quadrupole magnet. The size of the magnetic core is 176 mm × 176 mm. The B-H curve of the magnetic core is again shown in Fig. 8. Since the magnetic field at the center gap is zero in a quadrupole magnet, magnetic field at the Bohr radius is evaluated with various coil current densities from 0.5 to 5.0 mega ampere per square meter. The error of the magnetic field obtained by the MMM with MSC is within sub percent as shown in Figures 11. The computing time for analyzing the quadrupole magnet with Intel Core i7-9700K CPU at 3.60 GHz is shown in Table 2.

Table 2 CPU time for analysis of quadrupole magnet.

elements		method	current density (MA/m^2)	CPU time (sec) mesh	CPU time (sec) solver
9,013,890		Opera3D	0.5	2189	1623
			10.0	2189	24549
(a)	1,008	MMM	0.5	< 1	5
			10.0	< 1	12
		MMM with MSC	0.5	< 1	36
			10.0	< 1	45
(b)	640	MMM	0.5	< 1	3
			10.0	< 1	4
		MMM with MSC	0.5	< 1	10
			10.0	< 1	10

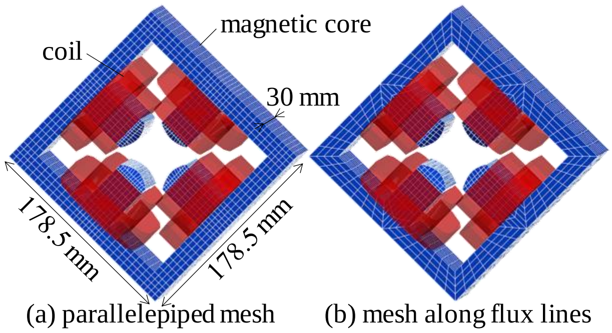


Fig. 10 Two different types of the mesh of the quadrupole magnet.

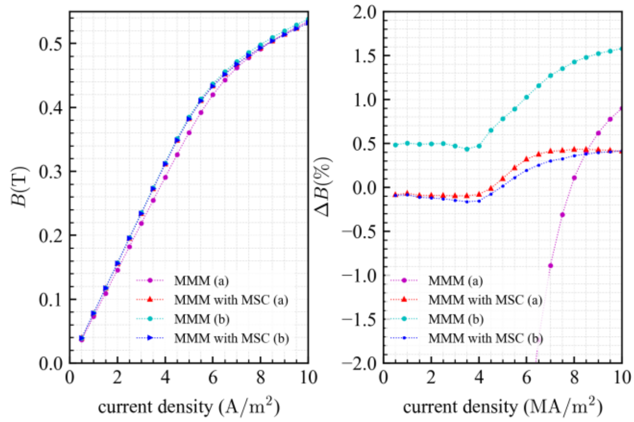


Fig. 11 Magnetic field and its errors at Bohr radius.

4. Conclusion

The Magnetic Moment Method with the idea of the Magnetic Surface Charge method is proposed that is free from the meshing problem. Unknown variables are surface magnetic charges on each face of the elements and the negative magnetic point charges are placed at the center of the gravity of the element. Accelerator magnets are analyzed, and it is shown that the proposed method outperforms currently available finite-element packages with respect to the CPU time and accuracy of the magnetic field. The proposed method is implemented in the commercial software ELF/MAGIC [10].

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